

Artefact Agreement

Team 224 and Queensland Rugby Union

1. PARTIES INVOLVED

This agreement is made between:

- Students from Queensland University of Technology (QUT) – Liam Mills, Elizaveta Zheltova, Brooke Hill, Upité Phiri, hereinafter referred to as ‘Party 1’
- Queensland Rugby Union Staff, hereinafter referred to as ‘Party 2’
- QUT Staff, Dimitri Perrin and Wayne Wood hereinafter referred to as ‘Party 3’

2. PURPOSE

The purpose of this agreement is to define the ownership, development, and usage terms of the artefact detailed below.

3. DESCRIPTION OF ARTEFACT

Title: Offside Detection Computer Vision System

Type: Software program

Brief Description: The Offside Detection Computer Vision System, hereinafter referred to as ‘the artefact,’ is a computer vision system which detects offsides within a Rugby Union game, with a focus on rucks and lineouts. The purpose of the artefact is to detect offsides to aid the Queensland Reds and wider Rugby Union stakeholders in offside awareness for coaches, players, teams and referees.

Scope: The scope of this project and the artefact is to output two offside detection modules, a ruck module and a lineout module, which will use video input to detect and visualize offsides. A database and Graphical User Interface (GUI) components are not included in the scope of the artefact.

4. ROLES AND RESPONSIBILITIES

Each party agrees to the following responsibilities:

Party 1:

- **Project Planning:** detailed planning and execution of the project and its milestones
- **Communication:** clear and consistent communication on the progress of the project, any challenges and recommendations
- **Model Development & Training:** model creation and training of models using machine learning with adjustments to fit outcome requirements

- **Testing & Quality Control:** training the system to have a high degree of accuracy and testing systems on real data to maintain reliability
- **Post-Deployment Support:** provide instructions, documentation and access to deliverables and project files for handover to industry partners

Party 2:

- **Define Requirements:** clearly outline the project goals, desired outcomes and detailed specifications, including the system inputs and outputs.
- **Resources Provision:** provide necessary content and resources within a reasonable time. Content can include but is not limited to video or photo datasets, hardware specifications, or access to existing systems.
- **Timely Feedback:** offer constructive feedback to the development team ensuring they understand the vision for the project.
- **Decision-Making:** be available for decision-making throughout the project life cycle, to provide final approvals for designs, models, and system implementations.

Party 3:

- **Technical Feedback:** provide feedback and recommendations for improvement on the technical aspects of the project
- **Project Management Guidance:** provide guidance on the organizational and project management aspects of the project

DELIVERABLES – MINIMUM VIABLE PRODUCT FEATURES

Computer Vision

Project files including:

- Main file to run the artefact
 - Main file will utilise all other files to execute ruck and lineout detection
- YOLO Rugby Model (trained on Roboflow Dataset)
 - Lineout detection
 - Ruck detection
 - Ball detection to help determine when a ball is released from a lineout or a ruck
 - Hooker detection to determine centre point of lineout
- Pre-Trained YOLO Player Model
 - Detect players on the field using default YOLO v8 model
 - Ankle detection using pose estimation
- Ruck offside detection
 - After ruck is detected, check when ruck has finished or ball is released
 - Get last feet of the ruck by detecting outermost points
 - Run field line detection to establish field parameters
 - Get intersection of field lines and draw a line through last feet to get offside line
 - Detect player ankles and check who is offside
- Lineout offside detection
 - After lineout detected, check when lineout has finished or ball is released twice
 - Get lowest point of the hooker or lineout
 - Run field line and point detection to establish field parameters
 - Translate points to top-down view

- Get offside points (+/- 10m from centre point)
- Get intersection of field lines and draw line through offside points to get offside line
- Detect player ankles and check who is offside
- Helper functions
 - Separate functions that implement the features listed above

Technical Documentation

- **Testing Documentation:** a comprehensive list of tests conducted during building and testing processes.
- **Troubleshooting:** written instructions to troubleshoot common errors

User Guide

- **The Artefact:** in-person demonstration during hand-over and written instructions on how to use the artefact.
- **Roboflow:** for creating datasets for YOLO model training. Written instructions on how to add more images to the datasets and retrain the models.

Installation Instructions

We have been working on this project in a Windows environment. As the Queensland Rugby Union team works with Mac devices, the installation instructions will be provided for both Mac and Windows devices. This will include written instructions for installing all required modules, including but not limited to:

- Python
- Ultralytics - YOLO
- Matplotlib
- NumPy
- OpenCV
- Scikit-learn
- Pydantic
- Tkinter
- Zscores from (scipy.stats)

DELIVERABLES – ADDITIONALLY IMPLEMENTED FEATURES

- User Interface
 - Select file from personal device
 - Check file type to be a video
 - Select frame from video and get user to set a threshold value
- YOLO Rugby Model (trained on Roboflow Dataset)
 - Rugby model detects referees, however this feature has not been utilised within our project. Despite this, it is a valuable feature that can be integrated in later iterations of the project

HANDOVER

On the **2nd of June 2025 at 12pm***, Party 1 will hold a meeting with Party 2 at Ballymore Headquarters where project deliverables will be discussed and a demonstration conducted to help set up and handover the project.

- * = to be confirmed

SIGNATURES

By signing the below, the parties acknowledge their understanding and agreement to the terms outlined above

QUT Students:

Name:	Elizaveta Zheltova	Liam Mills	Brooke Hill	Upité Phiri
Signature:				

Queensland Rugby Union Staff:

Name: **Sam Cordingley**

Signature: 

Academic Supervisor

Name:	Dimitri Perrin	
Signature:		QUT Staff
Name:	Wayne Wood	
Signature:		